

6 - WEEK MENTOR - LED PROGRAM

Bioinformatics × Artificial Intelligence

From raw sequencing reads to deployed AI models — genomics, single-cell and spatial omics, protein structure prediction, generative drug design and reproducible cloud pipelines.



6

Weeks, mentor-led

6

Live industry projects

40+

Tools & frameworks

1

Deployed MVP capstone

WHAT THE PROGRAM COVERS

● Genomics & Variants

NGS QC, alignment, GATK / DeepVariant calling, structural variants and long-read phasing.

● Single-Cell & Spatial

Scanpy / Seurat workflows, scVI integration, trajectories, Visium and Xenium deconvolution.

● Structure & Drug Design

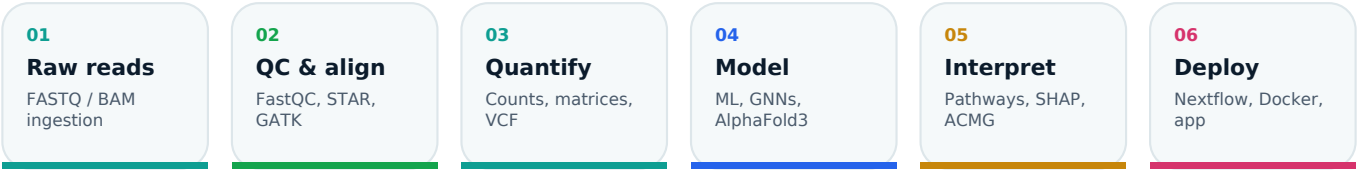
AlphaFold3, ESM-2 protein language models, docking, MD intuition and generative binders.

● ML, Graphs & MLOps

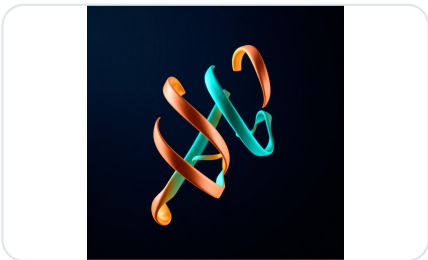
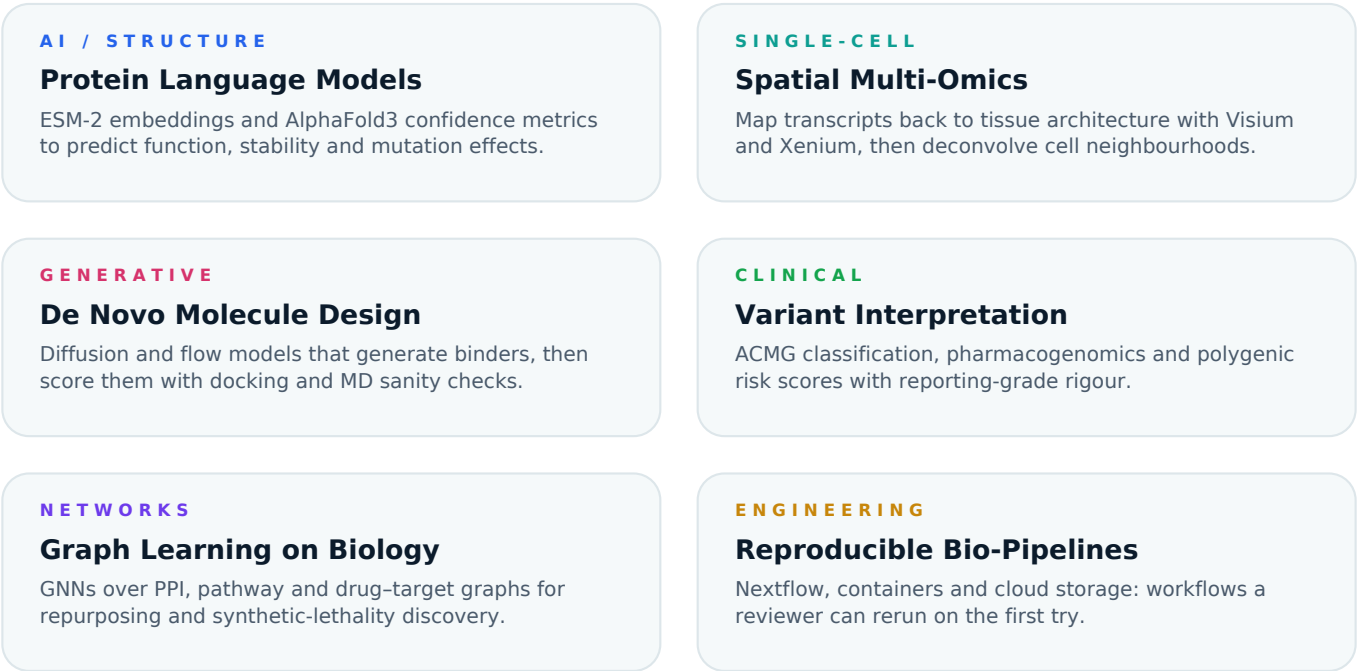
GNNs on biological networks, genomic transformers, GWAS/PRS, Nextflow and deployment.

THE END-TO-END PIPELINE YOU WILL BUILD

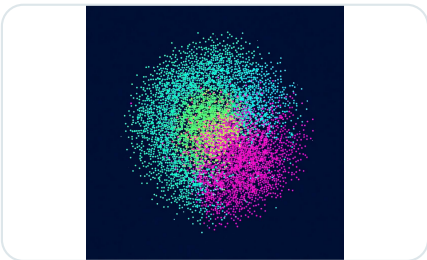
Raw reads to deployed model



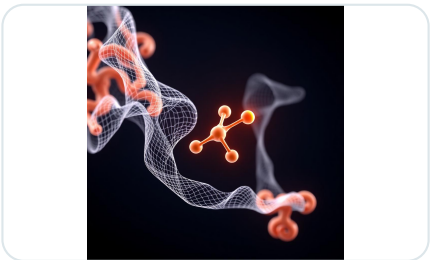
ADVANCED SPECIALISATION TRACKS



PREDICTED FOLDS



SINGLE-CELL UMAP



LIGAND DOCKING

Weeks 1 - 3

WEEK 01 FOUNDATION Bioinformatics Foundations & AI Integration TOPICS COVERED - Central dogma for data people: DNA → RNA → protein - Biological data formats: FASTA, GenBank, VCF, GFF, BAM/CRAM - NCBI, Ensembl, UniProt, PDB — programmatic access via REST APIs - Data cleaning, missing values, batch effects and outliers - Pairwise & multiple sequence alignment (global vs local) - Hidden Markov models and profile search (HMMER) TOOLS & FRAMEWORKS BLAST Clustal Omega MAFFT Biopython HMMER	WEEK 02 CORE Genomics, Variants & Transcriptomics TOPICS COVERED - Short vs long read sequencing: Illumina, PacBio HiFi, Nanopore - QC → alignment → variant calling (GATK best practices, DeepVariant) - RNA-Seq workflow: quantification, normalisation, DE testing - Pathway & gene-set enrichment: GO, KEGG, Reactome, GSEA - Structural variants, CNVs and long-read phasing - Cloud-scale genomics with cost-aware compute TOOLS & FRAMEWORKS FastQC STAR Salmon DESeq2 GATK DeepVariant	WEEK 03 ADVANCED Single-Cell & Spatial Omics TOPICS COVERED - scRNA-seq chemistry, droplet demultiplexing, ambient RNA removal - Dimensionality reduction: PCA → UMAP; clustering & marker discovery - Cell-type annotation, label transfer and reference mapping - Trajectory inference, RNA velocity and pseudotime - Spatial transcriptomics: Visium, Xenium, deconvolution - Batch integration with Harmony, scVI and scANVI TOOLS & FRAMEWORKS Scanpy Seurat scVI Squidpy CellRanger
Live Project 1 — Sequence retrieval, alignment & annotation pipeline	Live Project 2 — RNA-Seq differential expression & pathway report	Live Project 3 — Single-cell atlas: clustering, annotation & trajectory

Weeks 4 - 6

WEEK 04

ADVANCED

Structural Bioinformatics & AI Drug Discovery

TOPICS COVERED

- Protein structure hierarchy, folds and domain architecture
- AI structure prediction: AlphaFold3, ESMFold, RoseTTAFold
- Protein language models & embeddings for function prediction
- Molecular docking, scoring functions and virtual screening
- Molecular dynamics essentials and free-energy intuition
- Generative chemistry: diffusion models, de novo binder design

TOOLS & FRAMEWORKS

AlphaFold3

ESM-2

AutoDock Vina

RDKit

DeepChem

PyMOL

Live Project 4 — Fold prediction + docking & lead prioritisation

WEEK 05

FRONTIER

Machine Learning & Systems Biology

TOPICS COVERED

- Feature engineering on omics matrices; leakage & cross-validation traps
- Graph neural networks on PPI and pathway networks
- Transformers for genomic sequence (Enformer, DNABERT-style models)
- Multi-omics integration: MOFA, autoencoders, late fusion
- GWAS, polygenic risk scores, fine-mapping and Mendelian randomisation
- Explainability for clinical trust: SHAP, attention attribution

TOOLS & FRAMEWORKS

PyTorch

scikit-learn

PyG

PLINK 2

Cytoscape

MOFA+

Live Project 5 — Disease-risk model with GWAS features + explainability

WEEK 06

FRONTIER

Pipelines, MLOps & Translational Deployment

TOPICS COVERED

- Reproducible workflows with Nextflow / Snakemake + nf-core
- Containers, Conda envs and version-pinned reproducibility
- Cloud genomics on AWS/GCP; object storage & spot compute
- Experiment tracking, model registries and drift monitoring
- Metagenomics, pathogen surveillance and genomic epidemiology
- Ethics, consent, HIPAA/GDPR and clinical reporting standards

TOOLS & FRAMEWORKS

Nextflow

Docker

MLflow

Streamlit

Kraken2

Nextstrain

Live Project 6 — Deployed MVP: end-to-end pipeline + interactive app

OUTCOMES

What you walk away with

01

Read biology like data

Move fluently between raw reads, matrices, structures and networks without losing the biology.

02

Ship reproducible analysis

Every result you produce is containerised, versioned and rerunnable by someone else.

03

Apply AI where it earns its place

Know when a transformer beats a t-test — and how to validate either honestly.

04

Publish a portfolio MVP

A deployed, interactive bio-app with documented methods that recruiters can actually open.

WHO THIS IS FOR

- Life-science students & researchers
- Software engineers entering genomics
- MSc / PhD candidates needing compute skills
- Data analysts moving into biotech
- Pharma & clinical research associates
- Founders building health-tech products

Ready to start your bioinformatics career?

Mentor-led cohorts · 6 live projects · certificate & deployed capstone

edufyi@edufyitechsolutions.in



ACCREDITED CERTIFICATIONS.

Every learner receives industry-recognised, accredited certificates on completion of the internship, industrial training and capstone project modules.





INTERNSHIP CERTIFICATE

Accredited Certification





INDUSTRIAL TRAINING

Accredited Certification





PROJECT COMPLETION

Accredited Certification



INTERNSHIP

Verified internship completion credential



TRAINING

Industrial training hours certified



PROJECT

Capstone project completion proof

GLOBAL MNC CERTIFICATIONS.

Learners are prepared for and certified through globally recognised professional certification programs from leading technology companies.



CISCO
Cybersecurity CCST



MICROSOFT
Azure Fundamentals



IBM
Internet of Things



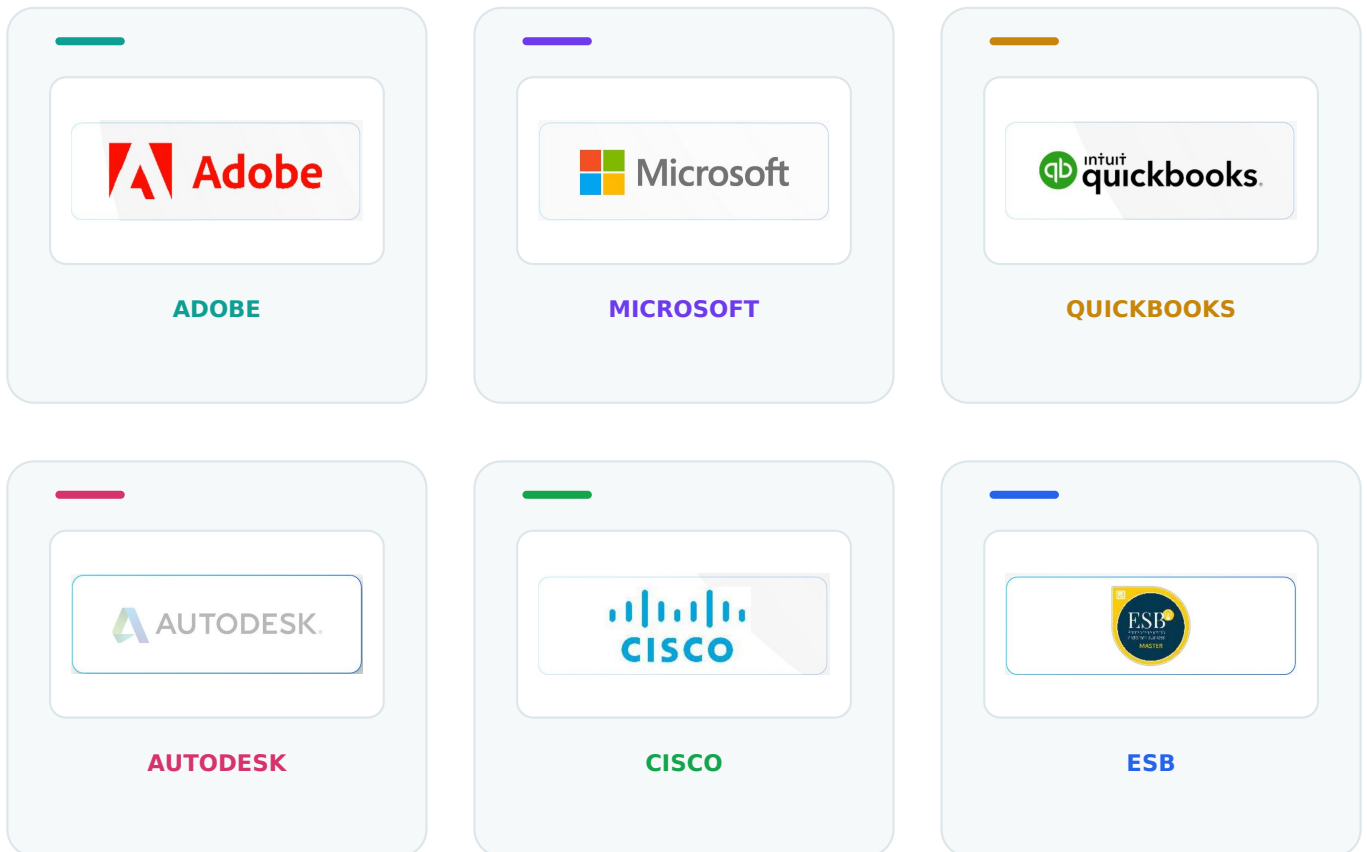
AUTODESK
Certified User



ESB
Entrepreneurship

AUTHORISED PARTNERS.

Edufyi Tech Solutions is an authorised certification and testing partner for the world's leading technology certification bodies.



AUTHORISED TESTING & CERTIFICATION CENTRE

Certify on campus with globally accepted credentials.

CERTIFY. BUILD. GET HIRED.



Accredited Certification



Microsoft Azure
Fundamentals



Cisco Cybersecurity
CCST



IBM Internet of Things



Adobe Certified
Professional



Autodesk Certified User



ESB Entrepreneurship



QuickBooks Certified
User

START YOUR JOURNEY

Enrol in the specialised Industrial Training & Internship program with 150+ partnered companies. Learn from industry experts, ship 6+ real-time projects and earn globally recognised certifications.

WEB www.edufytechsolutions.com

MAIL edufyi@edufytechsolutions.in

CALL +91 63620 00213

EDUFYI TECH SOLUTIONS

